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Thermal conductivity of thin single-crystalline germanium-on-insulator structures

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ABSTRACT

We report on the effective cross-plane thermal conductivity of single-crystal Ge layers from 42 to 100 nm on a SiO_2/Si substrate in the temperature range 30–300 K. We observe a drastic reduction of the thermal conductivity compared to bulk germanium. A large contribution to the temperature rise in the Ge layer is due to the interfacial thermal resistance between c-Ge/SiO_2. The use of a size-dependent intrinsic thermal conductivity of the Ge layer instead of the bulk thermal conductivity improves the consistency with values of the thermal boundary resistance derived from the diffusive mismatch model. Ultrathin films of Ge suffer from a lower reduction of the thermal conductivity compared to ultrathin films of Si, which makes germanium-on-insulator structures promising candidates for devices with reduced self-heating effects compared to silicon-on-insulator structures.

1. Introduction

The growth of thick, dislocation-free, pure single-crystalline Ge layers directly on Si is a challenge because of the difference in lattice parameter between Si and Ge. However, recent developments in wafer processing have permitted to achieve germanium-on-insulator (GeOI) structures with high-quality Ge layers up to 100 nm thick [1,2]. GeOI structures are potential candidates to improve transistor performance in CMOS microelectronics due to their superior charge carrier mobility compared to other semiconductors [3,4]. Because of their compatibility with III–V semiconductors such as GaAs, GeOI structures may also find applications in areas such as light detectors or emitters. Germanium has traditionally been disregarded in the microelectronics industry due to the lack of a suitable oxide and its non-compatibility with wet processes in CMOS technology. However, the introduction of high-K dielectrics opens up a possible use for Ge in large scale applications [5]. Surface passivation with a thin silicon cap [6] or with a germanium silicon-oxynitride has also been demonstrated [1].

With the shrinking of the dimensions to the nm-scale heat removal is becoming a major issue in the design of reliable devices. The GeOI structure exhibits nearly bulk electrical conductivity, high mechanical strength and on-insulator advantages with substantial benefits for microelectronic applications. Based on these properties, GeOI could provide a solution for the highly desirable on-chip integration of electronic and optoelectronic operations. However, the reduced thermal conductivity of bulk Ge compared to bulk Si may induce self-heating in the circuits that can adversely affect the operation of analog circuits. Therefore an accurate knowledge of the temperature rise is essential to assess the lifetime and reliability of devices [7]. Although the thermal properties of silicon-on-insulator wafers have been already analyzed, to our knowledge, no such measurements have been performed on GeOI structures. SOI structures 20–100 nm thick exhibit a significant reduction in thermal conductivity compared to bulk Si [8–12]. Previous measurements on SOI have shown that if the mean free path of the phonons is larger than the thickness of the film, the phonon propagation will be primarily limited by the boundary between the two materials and ballistic transport dominates heat conduction at the nm length scale [8–10].

In this article, we concentrate on the thermal properties of thin films of high-quality single crystalline Ge on an insulating SiO_2 layer. We measure the temperature-dependent effective cross-plane thermal conductivities of GeOI ultrathin films from 42 to 100 nm. The measured thermal conductivities of the 42 nm and 60 nm Ge layers are a factor of 5 and 3 smaller, respectively, than the bulk value of Ge at room temperature. The thinnest Ge films show a reduction of the thermal conductivity with decreasing thickness compared to silicon-on-insulator structures and may become more attractive than ultrathin silicon layers for large-scale integration of field effect transistors. We also derive expected values for the intrinsic thermal conductivity of the Ge layer by using a moment expansion of the Boltzmann equation that yields an approximate analytical expression for the thermal conductivity of nanosystems in terms of their size [13,14].

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Nomenclature

κ_{bulk}	bulk thermal conductivity (W/mK)
κ_{eff}	effective thermal conductivity (W/mK)
$\kappa_{film\ Ge}$	out-of-plane intrinsic thermal conductivity of the Ge layer (W/mK)
l	mean free path (nm)
L	Ge layer thickness (nm)
R_{th}	thermal resistance of interfaces (m^2 K/W)

2. Experimental procedure

The samples were GeOI structures from SOITEC produced by the SmartCut™ technology [2]. The starting Ge thickness was 100 nm on a buried oxide layer of 300 nm over a Si wafer of 750 μm thickness. The threading dislocation density is of the order of 10^6 cm^{-2} and the SiO_2/Ge interface and the Ge surface are atomically flat. The final thickness was achieved by successive oxidation steps in deionized water followed by rinses in warm water to remove the GeO_x layer. By this procedure we obtained single-crystalline Ge layers of 42, 60, and 100 nm thickness. Lower thicknesses were not attempted because, below 40 nm, the interface thermal resistance clearly dominates the measured thermal conductivity. The Ge thickness was evaluated by ellipsometry and the crystalline perfection of the layers was assessed by X-ray diffraction through rocking curves. The full-width-at-half-maximum of the rocking curve was 15 arcsec demonstrating the high crystallinity of the layer even after the etching process.

The cross-plane thermal conductivity was measured by the differential 3ω method developed by Cahill and Lee [15]. We used 100 nm thick Pt wires with a 5 nm Cr adhesion layer. A 200 nm plasma-enhanced chemically vapor-deposited SiO_2 layer was grown on the Ge samples, bulk single-crystal and GeOI thin films, and on a reference wafer (silicon substrate + buried oxide) to electrically insulate the heater/sensor. The thermal contribution of the Plasma-Enhanced Chemical Vapor Deposition (PECVD) SiO_2 layer plus the buried oxide plus the Si substrate is measured independently and subtracted from the measurements on the SiO_2/GeOI structure. All measurements were carried out between 30 and 300 K in a vacuum of 2.5×10^{-7} mbar and are plotted in Fig. 1. As apparent from the figure the slope of the four set of data is similar within 10% which indicates the good reproducibility between samples of the SiO_2 + substrate contribution to the measured signal. Joule heating was produced by applying an AC current of 15 mA using a Keithley 6221 source on the current pads of the sensor. The 3ω voltage was measured using a Stanford Research 830 lock-in amplifier. The thermal boundary resistance at the Ge/SiO_2 interface adds a frequency-independent component to the total temperature rise and therefore cannot be distinguished from the film thermal resistance. We apply the one dimensional (1D) solution of the heat conduction equation to obtain the thermal conductivity from the measured temperature rise. As reviewed recently by Tong and Majumdar [16] the temperature change across the film can be considered 1D in the limit of low film thermal conductivity compared to the substrate, large interface thermal conductance, and heater lines wider than the film thickness. In this work the heater-width-to-film thickness and the film/substrate thermal conductivity ratios are 150 and about 0.05–0.15 at 300 K, respectively. Based on these values the heat spreading in the parallel direction of the film should be small and the cross-plane thermal conductivity of the film can be calculated from the one-dimensional steady-state heat conduction model. Because the presence of the buried oxide layer introduces some uncertainty in the validity of this assumption we used a two-dimensional (2-D) finite-element thermal modeling to analyze the response of the system to a sinusoidal temperature gradient [17]. Fig. 2 shows the results of the simulation that demonstrate the heat flow through the structure is mainly one-dimensional in spite of the presence of the less conductive buried oxide between the Ge ultrathin film and the Si substrate. The thermal conductivity that is deduced from the temperature rise associated to the Ge film is an effective value that is subsequently analyzed beyond the Fourier approximation by using a moment expansion of the Boltzmann equation [14] and considering the contribution of the interface thermal resistance between c-Ge and SiO_2 in the frame of the diffusive mismatch model (DMM) proposed by Swartz and Pohl [18].

3. Results and discussion

Fig. 3 shows the measured effective out-of-plane thermal conductivity as a function of temperature for GeOI layers of thickness 42, 60 and 100 nm. The uncertainty of the measurement was determined taking into consideration the sources of noise (Johnson, shot, 1/*f*), the lock-in input noise that may interfere in the 3ω measurement, and the uncertainties associated with the thickness and dimensions of the heater. For clarity, only the error bars associated

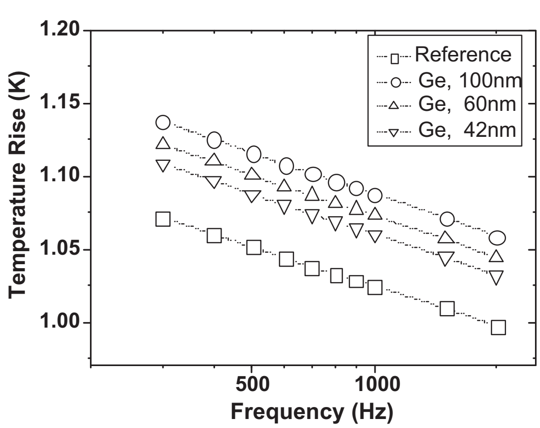

Fig. 1. Temperature rise as a function of heater frequency for the reference sample (silicon + buried oxide) and for Ge layers of thickness, 42, 60 and 100 nm.

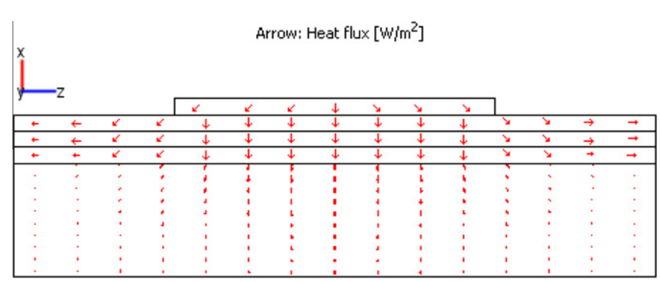

Fig. 2. Schematic view of the 2D heat flow diagram across the GeOI structure imposed by a temperature gradient generated by circulating a sinusoidal current through the metallic heater.

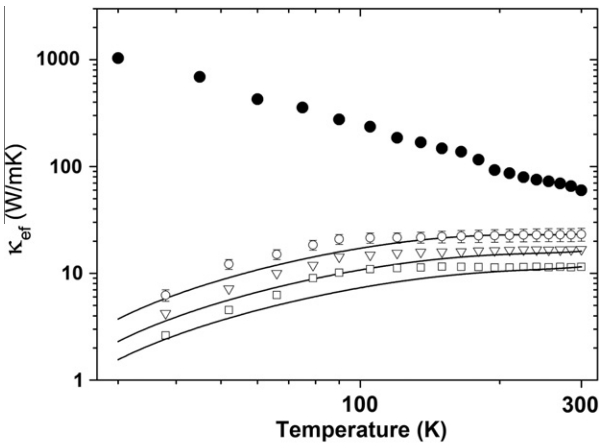

Fig. 3. Effective thermal conductivity for the different layers as a function of temperature (circles, 100 nm layer; downward triangles, 60 nm; squares, 42 nm). Solid lines are the values predicted by Eq. (2) with a thermal boundary resistance given by the diffusive mismatch model. Solid circles correspond to the thermal conductivity measured for bulk Ge.

with the GeOI sample of 100 nm are shown in the graph. For comparison, the measured data for bulk germanium, which closely agrees with literature values [19], is included in the figure (upper curve). A drastic reduction in thermal conductivity compared to the bulk value of Ge is observed for all thicknesses. Interestingly, there is also a dramatic difference in the thermal conductivity behavior with temperature between the bulk Ge and the thin films, as has also been reported for comparable SOI structures [9,10]. Whereas bulk Ge has maximum conductivity at 10–15 K and a value of ~60 W/mK at 300 K, the thin layers fabricated in this study show a maximum thermal conductivity at a much higher temperature (~200 to 250 K), and a value around 10 W/mK at 300 K for the 42 nm ultrathin film. At first view it may be surprising that the 100 nm film exhibit a steady increase up to 100 K followed by a plateau till 300 K, instead of a behavior closer to the bulk single-crystalline material. The likely explanation to this tendency may be related to the strong variation of the mean free path (MFP) with temperature [20]. At low T the mean free path is large because of the low phonon population and the small number of scattering events, however at high T the MFP is strongly reduced mainly by the existence of phonon–phonon Umklapp scattering. Following data from Chen, l_{Ge} in the Debye approximation is approximately 27 nm at room temperature [21], whereas at 100 K, l_{Ge} is around 130 nm. Therefore at low temperature in the sub-100 nm films phonon propagation approaches the ballistic regime and the impact of the interface thermal resistance is much stronger. In this regime, the measured out-of-plane effective conductivity can deviate significantly from the intrinsic thermal conductivity of the film due to the existence of the interface thermal resistance between the Ge layer and the oxide.

The data shown in Fig. 3 cannot be explained by using a simple expression as $\kappa_{ef}^{-1} = \kappa_{bulk}^{-1} + (R_{th}/L)$, where κ_{ef} is the effective thermal conductivity of the layer and κ_{bulk} that of the bulk, R_{th} the thermal resistance of the interface and L the thickness of the Ge layer. For a given temperature, a variation of R_{th} more than an order of magnitude would be required to fit the experimental data. This is unrealistic since all samples were prepared under identical conditions. In addition, the temperature variation requires an additional mechanism to explain the observed changes.

In an effort to isolate the contribution of the intrinsic conductivity of the Ge film to the measured values, we apply the model proposed earlier by Alvarez and Jou [13,14]. This model computes the intrinsic thermal conductivity from a modification of the heat transport equation, based on extended irreversible thermodynamics (EIT) [22]. This analysis has been shown to be very effective in predicting the thermal conductivity of Si nanosystems in terms of their size [23]. The analytical equation derived from this model is

$$\kappa_{film\,Ge} = \frac{\kappa_{bulk} L^2}{2\pi^2 \ell^2}\left(\sqrt{1+4\left(\frac{\pi\ell}{L}\right)^2} - 1\right) \tag{1}$$

where $\kappa_{film\,Ge}$ refers to the out-of-plane intrinsic thermal conductivity of the Ge layer, κ_{bulk} is the material bulk thermal conductivity, ℓ is the mean free path of the phonons, and L is the thickness of the layer. With this approach, we calculate the effective thermal conductivity of the GeOI structure incorporating the thermal boundary resistances at the Ge/SiO_2 interfaces as written in Eq. (2).

$$\kappa_{ef}^{-1} = \kappa_{film\,Ge}^{-1} + (R_{th}/L) \tag{2}$$

where the first term in the right-hand side is not the bulk conductivity, but a size-dependent conductivity given by (1). As when calculating conductivity for SOI structures, we assume that phonon scattering at the Ge/oxide interface is fully diffusive and therefore we calculate the thermal boundary resistance using the diffusive mismatch model of Swartz and Pohl [18] where scattered phonons lose their memory after reaching the interface. The thermal boundary resistance for the SiO_2/Ge/SiO_2 interfaces at 300 K is $2.2 \times 10^{-9}\ m^2$ K/W and increases to close to $1.5 \times 10^{-8}\ m^2$ K/W at 50 K. The results of the model derived from Eqs. (1) and (2) are shown as continuous lines in Fig. 3. A satisfactory agreement with the experimental data is achieved above 150 K. At low temperatures, however, predictions from the model underestimate the effective thermal conductivity, mainly because of the temperature dependence introduced by the diffusive mismatch model. Recent calculations by Mahajan et al. [24] using non-equilibrium molecular dynamics give values of the thermal boundary resistance between Si and SiO_2 in agreement with the DMM calculations. However, calculated room temperature values of the thermal resistance are typically one order of magnitude less than experimentally measured values. This is indeed what has been reported so far for Si/SiO_2, where R_{th} lies in the range of $2–6 \times 10^{-8}\ m^2$ K/W [25] and R_{DMM} equals $3.5 \times 10^{-9}\ m^2$ K/W. Ge has a lower acoustic mismatch with SiO_2 than Si; thus the thermal boundary resistance of the Ge/SiO_2 interface should be lower than that of Si/SiO_2 for a comparable interface quality. However, this difference will only account for a factor of two, which is insufficient to explain why the thermal resistance is so small. This discrepancy arises from our implicit assumption that part of the temperature rise in the GeOI structure

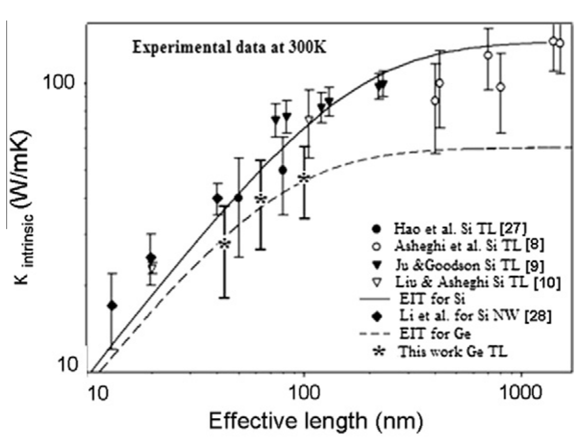

Fig. 4. Intrinsic thermal conductivity for silicon and germanium films as a function of the effective length (layer thickness or nanowire diameter). Symbols represent experimental values as noted in the legend. The solid line is the predicted curve of Eq. (1) for Si films with bulk value κ_0 = 140 W/mK and ℓ = 43 nm. The dotted line is the prediction for Ge films with bulk value κ_0 = 60 W/mK and ℓ = 20 nm.

is due to size effects in the thin Ge layer and not only to the boundary effect as stated in Eq. (1). Thus, the impact of the thermal boundary resistance in the overall temperature increase is reduced and can be reconciled with the values typically derived from DMM.

Finally, Fig. 4 plots the intrinsic out-of-plane thermal conductivity variation with thickness at 300 K compared to reported values for the in-plane behavior in comparable silicon-on-insulator structures [8–10,27,28]. The solid and dotted curves in the figure represent, respectively, silicon and germanium predictions of Eq. (1). Although in-plane and out-of-plane values differ, the ratio κ_{out}/κ_{in} should be constant in the reduced thickness range of our experiments. The intrinsic thermal conductivity of the 100 nm Ge layer at 300 K is 46 W/mK which approaches ($\kappa_{film}/\kappa_{bulk} = 0.8$) the value of bulk Ge, which is coherent with the high thickness of the layer compared with the mean free path at room temperature. The ratio is reduced to 0.25 for the 42 nm film. In spite of the reduced thermal conductivity of bulk Ge compared to Si ($\kappa_{Ge}/\kappa_{Si} \sim 0.4$), ultra-thin films of Ge undergo a significative variation of the thermal conductivity due to the small mean free path, as observed in Fig. 4, where the ratio κ_{Ge}/κ_{Si} for a 20 nm film is $\sim$0.8. This is in qualitative agreement with recent calculations in ultra-thin Ge films [26]. For the thinnest films, which are of interest for MOSFET devices, the GeOI structure should exhibit higher carrier mobilities. GeOI is therefore expected to provide improved performance compared to current SOI technology.

4. Conclusions

In conclusion, we have provided data on the thermal conductivity of single-crystalline Ge layers 40–100 nm thick grown on SiO_2/Si substrates. The effective cross-plane thermal conductivity of ultrathin germanium-on-insulator is drastically reduced compared to bulk Ge. The data have been interpreted by combining an intrinsic out-of-plane size-dependent thermal conductivity of the Ge layer and a thermal boundary resistance. The value of the latter, $R_{th} = 2.2 \times 10^{-9}$ m^2 K/W at 300 K, is consistent with the predictions of the diffusive mismatch model. The significative reduction of the effective thermal conductivity with decreasing thickness compared to silicon is beneficial for a potential large-scale integration of new transistors based on GeOI structures.

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